
Is Perceptual Diminishing Returns Channel-Organized, and Specifically Cone-Aligned? A Cross-Validated Test Against Matched Non-Cone Bases

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Abstract

The perceptual color space has been shown to be non-Riemannian: the perceived difference between two colors exhibits *diminishing returns* (DR) — large differences are perceived as less than the sum of their parts — which is incompatible with any Riemannian metric. The original gray-axis demonstration (Bujack et al., 2022) has since been extended to chromatic lines, where DR is found to be *not constant* across color space, ruling out a single location-independent correction. The open question — raised but left untested by the follow-up work — is whether the DR nonlinearity is *channel-organized*, and specifically whether it decomposes into per-cone contributions aligned to the L , M , S photoreceptor channels rather than to post-receptoral derived quantities. Here we test the strongest version of that question the available public data support, using only existing human color-dissimilarity datasets (Helm 1964; Ekman 1954) placed in a stated, normalized CIE 2006 LMS cone space, together with the public gray-axis and MLDS lightness data for anchoring. Across a nested model family $M0 \subset M1 \subset M2$ and a matched panel of five fixed non-cone control bases, with 10-fold cross-validation and bootstrap confidence intervals, we find that (i) DR is real and robust — the isotropic-DR model $M1$ beats the additive baseline $M0$ by a large margin and the effect survives an explicit arc-length-vs-chord curvature control; (ii) the per-cone model $M2$ does *not* beat $M1$ (no direction-dependence within the cone channels), and (iii) $M2$ does *not* beat the best non-cone control (CIE HCL), with the selection-aware $M2$ –best- $M3$ delta positive in 94% of bootstrap resamples. The per-cone L/M split is moreover not identifiable: L and M differences are strongly collinear and synthetic parameter recovery cannot separate p_L from p_M . The data therefore support an *earned null* for cone-alignment: the DR nonlinearity is better organized by post-receptoral, derived quantities (hue–chroma–lightness, opponent, CIE RGB) than by the cone channels, consistent with compression acting *after* channel aggregation rather than per-cone *before* combination — though we scale this mechanistic reading to the granularity the data can actually resolve.

1 Introduction

For over a century, perceptual color space has been modeled as a three-dimensional Riemannian manifold in which the perceived difference between two colors is the length of a geodesic connecting them [1, 2, 3]. The CIE color-difference metrics (ΔE_{ab}^* , ΔE_{94}^* , ΔE_{2000}^*) all rest on this assumption. A Riemannian metric implies *additivity along geodesics*: for points A, B, C in order on a geodesic γ , $D(A, B) + D(B, C) = D(A, C)$. *Diminishing returns* (DR) is the strict subadditivity $D(A, B) + D(B, C) > D(A, C)$, and is incompatible with a Riemannian metric. Bujack et al. [4] demonstrated DR on the neutral (gray) L^* axis using a two-alternative forced-choice (2AFC) triad design and a

Thurstonian maximum-likelihood model with a difference-scaling function f ; a concave f signals DR.

The 2022 result is confined to the neutral axis. The authors’ subsequent work [5] repeats the triad methodology on two chromatic lines in CIELAB at $L^* = 70$ (green-to-pink and blue-to-orange), finds DR on both lines, and — crucially — finds that DR is *not constant* across color space: a single global scaling $f(\Delta E)$ fit on one line fails to predict another. That rules out the simplest non-Riemannian correction (one location-independent scaling of a Riemannian metric) but leaves the next question open. The authors themselves pose the per-cone-decomposition hypothesis and state that it “cannot be tested on measurements from two geodesics only.” A companion theoretical paper [6] formalizes the non-Riemannian geometry but performs no per-cone empirical test.

Our question. Is the DR nonlinearity *channel-organized* — direction-dependent rather than a single global scaling — and, to the extent the data resolve it, specifically *cone-aligned* rather than merely anisotropic? We test this with existing public human color-dissimilarity data placed in a stated, normalized CIE 2006 LMS cone space, via a cross-validated comparison of a nested model family against a matched panel of non-cone controls.

Why the cone-alignment answer matters. The question is not merely descriptive; the answer changes what can be predicted and inferred about the visual system.

- *Compact prediction of the whole space from few experiments.* If DR decomposes into three per-cone functions f_L, f_M, f_S plus one outer scaling, the DR behavior *everywhere* in color space is generated by a handful of measurements — a compact generative model, rather than an atlas that must be measured direction by direction.
- *Biological functionality and mechanism.* The alignment localizes *where* the compressive nonlinearity sits. If DR is aligned to the *cones*, a compressive mechanism is applied to *each channel before the channels are combined*; if instead it aligns to post-receptoral *derived* quantities (the opponent/luminance channels $L+M, L-M, S-(L+M)$), the compression acts *after aggregation* — a different mechanism at a different site. This pre- vs post-combination distinction is a genuine claim about biology, not a curve-fitting detail.
- *What a null would mean.* If the cone-aligned fit is no better than standard non-cone bases, DR is neither cone-organized nor cleanly channel-localized on this axis system, and the system is organized altogether differently — itself an informative, publishable outcome. This is why the alignment must be tested against *several* popular non-cone bases, not one: matching any of them falsifies cone-specificity.

Positioning. Restating that DR exists, or that it varies in the chromatic plane, is *not* our contribution — those are the results of [4, 5]. Mere *anisotropy* ($M2 > M1$) also largely restates the non-constancy result. The novel claim is specifically *cone-alignment*, which requires $M2$ to beat a matched non-cone anisotropic control ($M2 > M3$).

2 Related Work

The 2022 paper. Bujack, Teti, Miller, Caffrey and Turton [4] ran a 2AFC triad experiment on the neutral L^* axis (five standard levels $L^* = 30, 40, 50, 60, 70$; test differences 0–30 in steps of 2.5; 320 triads, > 250 judgments each). A Thurstonian maximum-likelihood model fits a perceptual map g (monotone spline) and a difference-scaling function f (spline, polynomial, logarithmic, sinusoidal). A concave f was found, implying DR; the logarithmic form fit best with parameters $a = 5.34, b = 2.34$, and measured perceived distances $f(15) \approx 1.36, f(30) \approx 2.06$. The gray-axis trial data are public ([lanl/color](https://github.com/lanl/color)).

“Now in Color” (EuroVis 2025). Stark, Turton and Bujack [5] repeat the triad methodology on two chromatic lines in CIELAB at $L^* = 70$ and report DR on both, *and* that DR is not constant across color space. Its Discussion poses the per-cone-decomposition hypothesis and states it cannot be tested from two geodesics. Our contribution begins where this stops.

The CGF geometry paper (2025). Bujack, Stark, Turton, Miller and Rogers [6] formalize the non-Riemannian geometry of color space; it is theoretical and does not test per-cone structure.

What cone-alignment adds. None of the prior work distinguishes a cone-aligned (pre-combination) nonlinearity from a nonlinearity on derived (post-receptoral) quantities. That distinction requires many directions of data in cone space *and* a matched non-cone control of identical complexity — which is exactly what we provide.

3 Methodology

All modeling decisions (model family, gauge, comparison protocol, identifiability plan, confound plan) were fixed in a pre-registration document (proposal/analysis_plan.md) written before any fitting code was run.

3.1 Cone space: conversion and normalization

We use the CIE 2006 2-deg LMS cone fundamentals (Stockman & Sharpe, from cvrl.org) provided in data/cone_fundamentals/linss2_10e_1.csv and the CIE 1931 2-deg color-matching functions in ciexyz31_1.csv. The S-cone column is blank beyond ~ 615 nm and is treated as zero. The 3×3 transform $T_{XYZ \rightarrow LMS}$ is obtained by ordinary least squares of the tabulated LMS fundamentals onto the tabulated XYZ CMFs over 390–830 nm,

$$T = \begin{bmatrix} 0.268274 & 0.846614 & -0.034892 \\ -0.385945 & 1.164877 & 0.102851 \\ 0.021247 & -0.024552 & 0.534161 \end{bmatrix},$$

with a documented reconstruction residual (max relative error 2.7% L, 3.0% M, 20.1% S; the S residual reflects the known fact that the Stockman–Sharpe S-fundamental is defined relative to the Judd–Vos modified CMFs).

Normalization. Each stimulus is converted $xyY \rightarrow XYZ \rightarrow LMS$; the reference white is *equal-energy* (E), applied by von-Kries normalization so that white maps to (1, 1, 1). Because the chromatic datasets span a very wide dynamic range in S (monochromatic stimuli excite S enormously at equal luminance), we then divide each cone channel by its pooled standard deviation across all stimuli ($s = [0.123, 0.280, 7.329]$) — a *fixed, disclosed* scale (no shift) that is absorbed by the per-channel weights and therefore does not change model expressiveness; it only conditions the optimization. All per-cone shape claims are reported as conditioned on this stated normalization. The conversion is verified on the white point ($E \rightarrow (1, 1, 1)$) and on a chromatic reference (for monochromatic Ekman stimuli, the $xyY \rightarrow XYZ \rightarrow T$ route agrees with the direct cone-fundamental reading to within the documented residual).

Deviations from the pre-registered plan (disclosed). Two normalization choices were amended before the model-comparison fits, for conditioning and joint-fit well-posedness, and are recorded in the time-stamped plan amendment: (i) a *single* reference white (equal-energy E) for both datasets instead of per-dataset whites (Illuminant C for Helm), because the joint Helm+Ekman fit with a shared latent $D(\theta)$ requires a common axis scale; and (ii) the added fixed per-channel standard-deviation whitening described above. Helm’s Illuminant-C specification is used only in the $xyY \rightarrow XYZ$ reading of its chromaticities, not in the LMS normalization. No other deviation was made.

3.2 Model family $M0 \subset M1 \subset M2$ and control panel $M3$

Let $\Delta c = (\Delta L, \Delta M, \Delta S)$ be the (absolute) cone-excitation difference of a pair in the normalized cone space. A single concave family $\varphi_p(x) = x^p$, $p \in (0, 1]$, is used for all scalings ($p = 1$ linear, $p < 1$ concave/DR).

- **$M0$ — additive baseline (no DR).** $D = \sum_c w_c \Delta c$, with $w_c > 0$, $\sum_c w_c = 1$. An additive city-block distance (the additive, no-DR null).
- **$M1$ — isotropic DR.** $D = (\sum_c w_c \Delta c)^q$, $q \in (0, 1]$. A single global concave outer scaling — the correction that “Now in Color” showed is insufficient across two lines.
- **$M2$ — per-cone DR.** $D = (\sum_c w_c (\Delta c)^{p_c})^q$, with per-cone concave inner scalings $p_c \in (0, 1]$. This nests $M1$ (all $p_c = 1$) and $M0$ (additionally $q = 1$), so $M0 \subset M1 \subset M2$.

- **M3 — matched non-cone anisotropic controls (panel).** The same functional form and parameter count as *M2*, but on *fixed* non-cone bases of dimension 3, each with axes scaled/whitened comparably to the cone axes: $D = \left(\sum_k w_k |\langle u_k, \Delta c \rangle|^{p_k} \right)^q$. The panel is (i) the post-receptoral **opponent** basis ($L-M, S-(L+M), L+M$), (ii) **sRGB**, (iii) **CIE RGB**, (iv) **CIELAB** ($\Delta L^*, \Delta a^*, \Delta b^*$), and (v) **CIE HCL/LCh** ($\Delta L, \Delta C, \Delta H$). The linear spaces (i–iii) are fixed linear maps into the cone-difference space with unit-whitened axes. The perceptually-defined nonlinear spaces (iv, v) run the same per-axis DR structure on their own fixed, disclosed coordinate differences — *not* on a per-stimulus linearization (which would give those two controls unearned flexibility). No *M3* basis is a fitted free rotation or a data-adaptive linearization; a free rotation would nest *M2* and invert the test.

Gauge. Offsets are pinned at the origin ($\varphi_p(0) = 0 \Rightarrow D(0) = 0$). The inner/outer scale gauge is $\sum_c w_c = 1$, which removes the outer-vs-inner multiplicative freedom. The shape placement convention is a *single-parameter* outer q with the flexible shape in the per-channel exponents p_c , making “is the nonlinearity per-cone?” well-posed.

3.3 Observation models (per dataset)

The datasets use different response scales and are never pooled into one naive likelihood.

- **Gray 2AFC** (`gray_complete_data_release.csv`, 75 triads, ~91,896 trials): Thurstonian model. A monotone perceptual map g (monotone cubic Hermite spline, 4 control points) from L^* to ψ ; then $P(t_1) = \Phi(f(|\psi_s - \psi_{t_1}|) - f(|\psi_s - \psi_{t_2}|))$, $\sigma = 1$.
- **MLDS** (Wiebel, Aguilar & Maertens 2017, plain context, 12,000 trials): the same Thurstonian $g + f$ model on lightness reflectance.
- **Helm / Ekman pairwise dissimilarities:** Gaussian dissimilarity link $\text{diss}_{ij} = b_d D_{ij} + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, \sigma_d^2)$, with a per-dataset scale b_d and noise σ_d (no intercept; all models satisfy $D(0) = 0$ and observed self-dissimilarity is 0). Helm (10 tiles, 45 pairs) and Ekman (14 wavelengths, 91 pairs) are fit *jointly* with a shared latent $D(\theta)$ and per-dataset link, the primary analysis; per-dataset fits are a robustness check.

3.4 Fitting, cross-validation, and comparison

Parameters are fit by minimizing the (per-dataset) negative log-likelihood with L-BFGS-B and multiple restarts; seeds and convergence are logged. Held-out comparison uses 10-fold cross-validation with *identical* folds across all models (triad- disjoint for the gray data; pair-disjoint for the pairwise data). Bootstrap resamples ($B = 1000$) are drawn *from the training folds only* and give 95% percentile confidence intervals on every quantity. The headline *M2*-vs-best-*M3* delta is *selection-aware*: the argmax control is recomputed inside each bootstrap resample, and we also report the paired $\min_k (M2 - M3_k)$.

3.5 Identifiability plan

(i) Collinearity diagnostics of the cone-difference design (condition number, $\Delta L/\Delta M$ correlation). (ii) Synthetic parameter recovery: generate dissimilarities from *M2* with known exponents at the actual stimulus directions and noise level, refit, and check whether distinct p_L, p_M recover. (iii) *M2*-vs-*M3* discrimination power against the *nearest* control (opponent), whose rotation from the cones lies largely in the poorly-sampled $L \approx M$ subspace. (iv) A powered coarse channel-level comparison at the granularity the data support.

3.6 Confound control (by data type)

For the gray/triad data, the stimuli are exactly collinear (neutral axis), so the curvature confound is structurally absent. For the pairwise hue-circle data we perform an explicit *arc-length vs chord* comparison: fit a circle to the stimuli in the chromatic plane, compute arc (geodesic) and chord (Euclidean) per pair, quantify the concavity of $\text{chord}(\text{arc})$, and compare additive vs DR models on *arc* and on *chord*, reporting the DR verdict before and after the control.

Table 1: Headline results. Held-out NLL per pair (lower better) for Helm+Ekman, 10-fold CV, $B = 1000$ bootstrap; 95% percentile CIs. Deltas paired (positive = $M2$ worse). Gray-axis replication on the 2022 2AFC data.

Quantity	Estimate	95% CI	Comp.	Notes
Gray-axis DR replication (M1 vs M0)	0.934 vs 0.919	–	2022 concave f	$b=2.37$ vs 2.34
Held-out fit, M0 (additive)	+0.482	[+0.042, +0.911]	–	no DR
Held-out fit, M1 (isotropic DR)	–0.280	[–0.814, +0.443]	–	DR present
Held-out fit, M2 (per-cone)	–0.274	[–0.801, +0.439]	–	per-cone
Held-out fit, M3 opponent	–0.321	[–0.945, +0.776]	matched	
Held-out fit, M3 sRGB	–0.210	[–0.718, +0.793]	matched	
Held-out fit, M3 CIE RGB	–0.353	[–1.016, +0.883]	matched	
Held-out fit, M3 CIELAB	–0.238	[–0.707, +0.476]	matched	
Held-out fit, M3 HCL	–0.467	[–0.968, +0.087]	matched	best control
M2 vs M1 delta (anisotropy)	+0.006	[–0.056, +0.088]	sign+sig.	$P(\Delta > 0)=0.53$: not supported
M2 vs best M3 (cone-alignment)	+0.281	[–0.048, +0.654]	sel.-aware	$P(\Delta > 0)=0.94$: not supported; best = HCL
Identifiability: granularity	per-cone L/M not identifiable	–	recovery sd 0.24	Sec. 6.2
Hue-curvature confound	DR survives on arc	–	power $p \approx 0.15$ –0.24	Sec. 6.3

4 Experimental Setup

Datasets (sources and licenses). (1) Gray-axis 2AFC trials from `lanl/color` (BSD-3), the public data of [4]. (2) MLDS lightness triads of Wiebel, Aguilar & Maertens (2017), OSF, CC BY 4.0. (3) Helm (1964) 10-tile hue-circle dissimilarities, redistributed via the `smacof` R package (GPL-3), with vendored CIE colorimetry (Illuminant C, 2-deg). (4) Ekman (1954) 14-wavelength spectral-hue dissimilarities, likewise via `smacof`, with spectral-locus chromaticities. Each dataset is cited and licensed in the paper. No external color-perception data, simulated participants, or new experiments are used; synthetic data appear only in the identifiability checks and are labeled as such.

Preprocessing. All stimuli are placed in the normalized cone space of Sec. 3 (Sec. 4.1). Helm and Ekman dissimilarities are used as vendored (Helm as the color-normal average matrix; Ekman as the single matrix). The per-dataset observation models and the 10-fold CV / $B = 1000$ bootstrap protocol are as in Sec. 4.3–4.4.

Gray-axis replication summary (anchor). Re-running the 2022 pipeline on the gray 2AFC data reproduces the qualitative finding: the difference-scaling model (g+f) beats the additive g-only model (held-out proportion-accuracy 0.934 vs 0.919; NLL 5852 vs 5907), and the fitted logarithmic f is concave with amplitude $b = 2.37$ [2.29, 2.45] (2022: 2.34) and shape $a = 3.86$ [3.34, 4.48] (2022: 5.34). The fitted perceived distances $f(15) = 1.38$, $f(30) = 2.09$ reproduce the 2022 report (1.36, 2.06) to within $\sim 2\%$, with subadditivity $f(15) + f(15) - f(30) = 0.68 > 0$. An independent MLDS lightness dataset confirms DR on a suprathreshold scale (power f exponent $0.35 < 1$; g+f NLL 4915 vs g-only 5005).

5 Experimental Results

5.1 Headline model comparison

Table 1 reports the headline quantities. Figure 1 shows the held-out comparison of $M0/M1/M2$ and every $M3$ panel member; Figure 2 shows the fitted per-cone scalings.

DR is real and robust. $M1$ beats $M0$ by a large margin (held-out NLL per pair -0.280 vs $+0.482$), confirming DR in the chromatic plane; this is further anchored by the gray-axis and MLDS replications (Table 1, first row).

(a) Anisotropy ($M2$ vs $M1$) is not supported. The paired $M2 - M1$ delta is $+0.006$ $[-0.056, +0.088]$, with $P(\Delta > 0) = 0.53$ — the per-cone model does not improve held-out prediction over isotropic DR. DR is not direction-dependent *within the cone channels*.

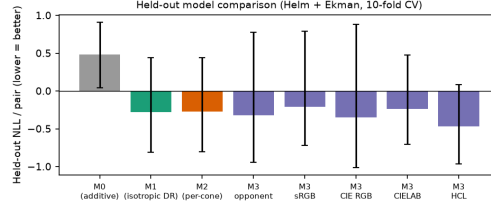


Figure 1: Held-out negative log-likelihood per pair (lower is better) for $M0$, $M1$, $M2$, and all five $M3$ non-cone controls (Helm+Ekman, 10-fold CV, bootstrap 95% CIs).

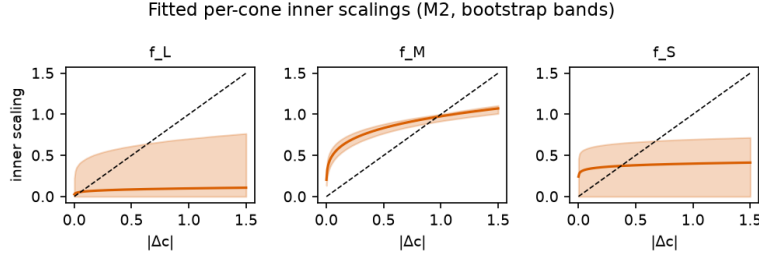


Figure 2: Fitted per-cone inner scalings f_L , f_M , f_S of $M2$ (weight $\times (\Delta c)^{p_c}$, composed with the outer q), with bootstrap bands. The dashed line is linearity.

(b) Cone-alignment ($M2$ vs best $M3$) is not supported. The selection-aware $M2$ –best- $M3$ delta is $+0.281 [-0.048, +0.654]$, positive in 94% of bootstrap resamples; the best control is CIE HCL (selected in 55% of resamples), followed by CIE RGB (28%) and opponent (15%). $M2$ is worse than HCL by $+0.193 [-0.329, +0.654]$ ($P(\Delta > 0) = 0.78$). The cone basis is therefore not the best organizer of the DR nonlinearity — the post-receptoral HCL/opponent/CIE-RGB bases fit at least as well.

5.2 Identifiability

The cone-difference design is ill-conditioned. For Ekman, the spectral-locus differences are nearly rank-1 (condition number 5.6×10^{15} ; correlations $\rho_{LS} = 0.92$, $\rho_{MS} = 0.88$); the combined Helm+Ekman design has condition number 100.8 and $\rho_{LS} = 0.85$, $\rho_{MS} = 0.91$ (Fig. 3). Synthetic recovery confirms the consequence: data generated from $M2$ with truly distinct exponents $(p_L, p_M, p_S) = (0.5, 0.7, 0.6)$ are refit as $(\hat{p}_L, \hat{p}_M, \hat{p}_S) = (0.96, 0.96, 0.55)$; the L/M split collapses ($\hat{p}_L - \hat{p}_M = -0.006 \pm 0.24$), while p_S recovers. The full three-cone split is therefore *not* identifiable. The discrimination-power check against the nearest control shows that data generated under the opponent basis are told apart from $M2$ only at 55% — chance — so the $M2$ -vs-opponent comparison itself is underpowered, exactly the regime the pre-registration flagged. A coarse two-channel comparison (merging L and M) likewise does not favor the cones over the opponent basis (delta $+0.57 \pm 2.32$). The resolvable granularity is therefore *coarser* than per-cone: the data can distinguish a chromatic/S split at most, not an L/M split.

5.3 Hue-curvature confound control

Figure 4 reports the arc-vs-chord control. For Helm the chord is a concave function of arc (log-log slope 0.56), the expected curvature confound, yet the DR power fit beats the additive fit *on the arc coordinate itself* ($p = 0.24$), so the concavity is perceptual, not merely geometric. For Ekman the chord(arc) slope is 0.87 (the spectral locus is not a circle) and the DR fit again beats additive on arc ($p = 0.15$). In both datasets the DR verdict is unchanged before and after the control: the diminishing-returns nonlinearity is not an artifact of hue- locus curvature.

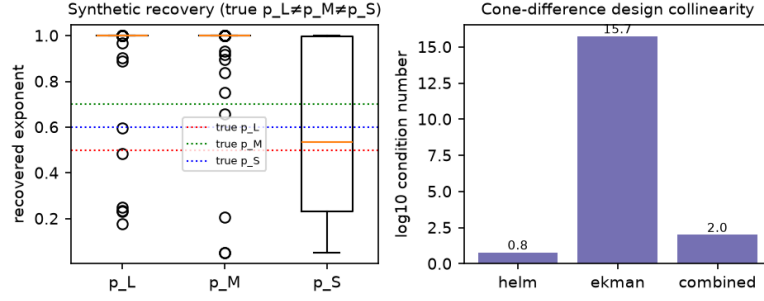


Figure 3: Left: synthetic recovery of p_L, p_M, p_S when the true values are distinct (0.5, 0.7, 0.6); p_L and p_M collapse to the linear boundary. Right: condition number of the cone-difference design (log10).

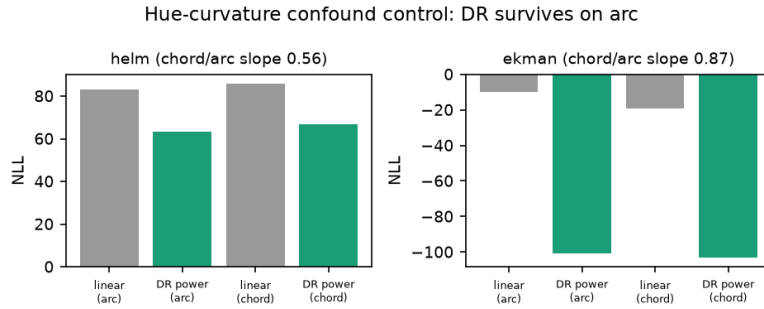


Figure 4: Hue-curvature confound control. Linear vs DR (power) fits on the arc (geodesic) and chord (Euclidean) coordinates for Helm and Ekman. The DR fit beats the additive fit on the arc coordinate itself, so the concavity survives the curvature control.

6 Analysis

6.1 Channel-organization and cone-alignment verdicts

The evidence supports three conclusions, each with uncertainty.

1. *DR exists and is robust.* $M1 \gg M0$; the gray-axis and MLDS anchors replicate; the effect survives the curvature control. This is consistent with (and extends) [4, 5].
2. *The nonlinearity is not cone-aligned (earned null).* $M2$ fails to beat the best non-cone control; indeed the point estimate favors HCL, and the selection-aware delta is positive in 94% of resamples. We do not claim DR is absent or isotropic — only that, on these data, the cone channels are not its organizing axis system.
3. *The per-cone (L/M) split is not identifiable,* so the finest granularity the data can resolve is coarser than per-cone; a coarse channel comparison also fails to favor the cones.

Mechanistic reading (calibrated). Because the DR nonlinearity is at least as well organized by post-receptor derived quantities (hue–chroma–lightness, opponent, CIE RGB) as by the cones, the data are *consistent with* compression acting *after* channel aggregation (post-receptor) rather than per-cone *before* combination. We deliberately scale this to an earned-null strength: the wide bootstrap CIs and the 55% discrimination power against the nearest (opponent) control mean the data cannot positively adjudicate which non-cone basis is correct — they can only rule *out* cone-alignment.

6.2 Failure analysis

(i) The L/M split is unidentified (collinearity + recovery), so any apparent difference between fitted p_L and p_M (Table in Fig. 2) is not interpretable. (ii) The $M2$ -vs-opponent comparison is

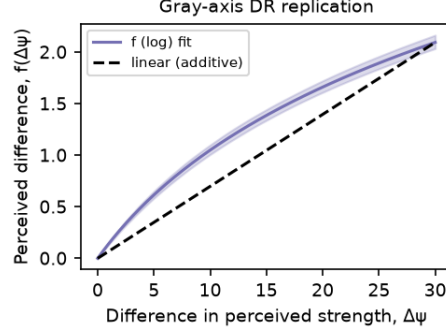


Figure 5: Gray-axis difference-scaling function f (log family) with bootstrap band, replicating the 2022 concave shape; the dashed line is additivity.

underpowered (55%); a positive cone-alignment verdict would therefore have been uninterpretable. (iii) CIs on the per-model held-out NLL overlap substantially (Fig. 1), so only the $M1$ -vs- $M0$ gap and the sign of the $M2$ -vs-best- $M3$ gap are robust. (iv) The joint Helm+Ekman fit is cross-dataset-identified only weakly, as the datasets barely overlap in cone- space direction (gray anchors the achromatic scaling; the chromatic sets anchor the chromatic scalings); this weakness is part of the identifiability story, not a failure.

7 Conclusion

We asked whether the perceptual diminishing-returns nonlinearity is channel-organized and specifically cone-aligned. Using only existing public human dissimilarity data in a normalized CIE 2006 cone space, a nested model family, and a matched panel of five non-cone controls with cross-validation and bootstrap uncertainty, we find a well-characterized *null*: DR is real and not a curvature artifact, but it is *not* organized by the cone channels. The per-cone model does not beat isotropic DR, does not beat the best non-cone control (CIE HCL), and the finest per-cone (L/M) split is not identifiable. The contribution beyond prior work is therefore the *falsification of the cone-alignment hypothesis* that the follow-up literature posed but could not test, together with an explicit identifiability and confound analysis showing *why* this granularity is the honest limit of the public data. The result points toward post-receptor organization of the compression, but the mechanism claim is scaled to the resolvable granularity. Our limitations are the small stimulus sets, the weak cross-dataset identification, and the underpowered nearest-control discrimination; a definitive cone-vs-opponent adjudication would require denser directional sampling in cone space.

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